

Jan Sobotka

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EDUCATION

Swiss Federal Institute of Technology in Lausanne (EPFL) <i>M.S. in Computer Science</i>	Sep. 2024 – Aug. 2026 <i>Lausanne, CH</i>
Czech Technical University, Faculty of Information Technology <i>B.S. in Informatics, Specialization in Artificial Intelligence</i>	July 2021 – June 2024 <i>Prague, CZ</i>

INDUSTRY EXPERIENCE

Research Data Scientist <i>AXA, Advanced AI Modeling Team</i>	Feb. 2026 – Aug. 2026 <i>Paris, FR</i>
- Built continued pre-training pipelines to study data impacts on the development of inductive reasoning and generalization in language models (70M–7B, FSDP). - Identified inverse scaling of inductive reasoning, where at a fixed data budget larger models shift from generalization toward memorization. Relevant for data curation and domain adaptation.	
AI/ML Engineer Junior <i>Generali Czech Republic</i>	June 2021 – June 2023 <i>Prague, CZ</i>
- Deployed machine learning solutions to production for text and product recommendation (35% improvement over previous method), client churn prediction, email classification, and customer segmentation. - Built computing infrastructure for the company's 90+ member internal data science community.	

PUBLICATIONS

* DENOTES EQUAL CONTRIBUTION

- [1] The Development of Property Induction During Language Model Pre-training.
Jan Sobotka, Ekaterina Goliakova, Xavier Renard, Antoine Bosselut.
In preparation.
- [2] Why Do LLMs Struggle in Strategic Play? Broken Links Between Observations, Beliefs, and Actions. [PDF] 📄
Jan Sobotka, Mustafa O. Karabag, Ufuk Topcu.
Under review (PNAS).
- [3] Do LLMs Strategically Reveal, Conceal, and Infer Information? A Theoretical and Empirical Analysis in The Chameleon Game. [PDF] 📄
Mustafa O. Karabag, **Jan Sobotka**, Ufuk Topcu.
Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026, Extended Abstract).
- [4] Reverse-Engineering Memory in DreamerV3: From Sparse Representations to Functional Circuits. [PDF] 📄
Jan Sobotka, Auke Ijspeert, Guillaume Bellegarda.
Neural Information Processing Systems (NeurIPS 2025, Spotlight at Mech Interp Workshop).
- [5] Weak-to-Strong Generalization under Distribution Shifts. [PDF] 📄
Jan Sobotka*, Myeongho Jeon*, Suhwan Choi*, Maria Brbić. (joint first authors reordered here)
Neural Information Processing Systems (NeurIPS 2025).
- [6] MEIcoder: Decoding Visual Stimuli from Neural Activity by Leveraging Most Exciting Inputs. [PDF] 📄
Jan Sobotka, Luca Baroni, Ján Antolík.
Neural Information Processing Systems (NeurIPS 2025).
- [7] Enhancing Fractional Gradient Descent with Learned Optimizers. [PDF] 📄
Jan Sobotka, Petr Šimánek, Pavel Kordík.
ArXiv preprint (2025).
- [8] Investigation into the Training Dynamics of Learned Optimizers. [PDF] 📄
Jan Sobotka, Petr Šimánek, Daniel Vařata.
International Conference on Agents and Artificial Intelligence (ICAART 2024).
- [9] Investigation into the Training Dynamics of Learned Optimizers (Student Abstract). [PDF] 📄
Jan Sobotka, Petr Šimánek.
AAAI Conference on Artificial Intelligence (AAAI 2024).

RESEARCH EXPERIENCE

- Research Assistant at the Autonomous Systems Group** July 2025 – Apr. 2026
University of Texas at Austin, Oden Institute for Computational Engineering and Sciences Austin, US
- Analyzed large language models in strategy games using mechanistic interpretability and game theory [1,2].
 - Supervised by Prof. Ufuk Topcu.
- Research Assistant at the MLBio Lab** Aug. 2024 – July 2025
Swiss Federal Institute of Technology in Lausanne (EPFL) Lausanne, CH
- Fine-tuned 70B+ multimodal foundation models (DeepSpeed, LoRA) for captioning and semantic segmentation.
 - Enhanced *Weak-to-Strong generalization* (learning from weaker teachers) by dynamically reweighting teaching signals, improving robustness under distribution shifts by over 30% [4].
 - Supervised by Prof. Maria Brbić.
- Research Intern at the Computational Systems Neuroscience Group** Sep. 2023 – Aug. 2024
Faculty of Mathematics and Physics, Charles University Prague, CZ
- Developed a state-of-the-art method for decoding images from neural population activity [5].
 - Supervised by Prof. Ján Antolík.
- Computational Neuroscience Research Intern** July 2023 – Sep. 2023
Biozentrum, University of Basel Basel, CH
- Designed computational models of spiking neural networks and applied dynamical systems theory to demonstrate how bistable neuronal dendrites enable a 1,000-fold increase in memory capacity.
 - Supervised by Dr. Everton J. Agnes.
- Research Assistant at the Data Science Lab** Apr. 2023 – Feb. 2024
Faculty of Information Technology, Czech Technical University Prague, CZ
- Analyzed training dynamics of *Learning-to-Optimize* (meta-learning) [7,8] and *fractional gradient descent* [6].

SELECTED PROJECTS

- Deep Reinforcement Learning for Optimal Experimental Design in Biology** Jan. 2023 – June 2023
- Open-science project focused on efficient estimation of biological system parameters [OpenBioML ].
- Generative Models of Regulatory DNA Sequences Based on Diffusion Models** July 2022 – Dec. 2022
- Open-science project investigating the application of diffusion models to genomics data [OpenBioML ].

EXTRACURRICULAR ACTIVITIES

- Organizer of the Traion Community of Student Entrepreneurs** June 2020 – Feb. 2021
- Organized offline meetings, educational seminars, and workshops for student entrepreneurs.
- Volunteer for an Entrepreneurship Education Program for Students** Dec. 2019 – Aug. 2020
- Organized events and wrote a technology/entrepreneurship blog for the Soutěž and Podnikej organization.
- Pitcher at the Czech Republic National Baseball Team U-15** Jan. 2017 – July 2017
- Secured third place at the U-15 European Baseball Championship 2017.

HONORS AND AWARDS

The Bakala Foundation Scholarship: Awarded to top 7% of applicants (academics & extracurriculars) | 2024
Merit-Based Scholarship, Czech Technical University: Awarded for a high academic average | 2021, 2022, 2023
National Benchmark Exam in Mathematics SCIO : Scored higher than 97% of the 875 test takers | 2021
Algorithms & Programming Competition FIKS: Ranked 4th out of 107 participants | 2020
TOP25 Czech High School Students of the Year 2020: Selected based on extracurricular activities | 2020

SKILLS

Programming languages: Python, C, C++, Julia, JavaScript, Go
Other selected technologies: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Azure, AWS, Git, Docker, Bash
Languages: Czech (native), English (C1, TOEFL iBT 105), German (A2)